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Recyclable cosmetic compact

Abstract

A tri-fold recyclable compact base including a pan receiving portion having an inner ring and an outer ring, the inner ring or the outer ring including at least one catch, and the inner ring configured to removably receive a cosmetics pan, a pan securing portion, the pan securing portion including a top flange and an inner ring extending perpendicularly from the top flange and an outer ring extending perpendicularly from the top flange, the inner ring or the outer ring including at least one circumferential ring configured to mate with the at least one catch, a first hinge rotatably connecting the pan receiving portion and the pan securing portion, a mirror receiving portion, and a second hinge rotatably connecting the mirror receiving portion and the pan securing portion.

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Background/Summary

FIELD

(1) This disclosure relates to a make-up case or a cosmetic compact. More particularly, the

disclosure relates to a cosmetic compact case that is refillable and can be broken down to maximize recyclable content.

BACKGROUND

(2) Conventional compact cases hold makeup or cosmetics, such as, powders, eye shadow, eyeliner, lipstick, or other beauty aids. Compacts often provide a mirror for viewing the face, thereby facilitating the process of applying; the makeup to the face.

(3) A compact usually includes a lid or a cover section hingeably coupled to a base or a container section. Either the cover section or the base section can be pivoted about the hinge to obtain the closed configuration of the compact, thereby providing a convenient storage device. The mirror is typically disposed on the inside of the compact cover section. The base section of conventional compacts usually includes a metal pan for containing the makeup. Alternatively, the makeup can be directly stored in a cavity in the base section.

(4) The mirror and the pan are typically glue to the respective inner surfaces of the lid and base. Because of their being glued to the respective inner surfaces the entire compact must be placed in normal trash and cannot be recycled. In the absence of the glue however, the pan and mirror are very difficult to reliably secure in the compact. Accordingly, improvements are needed to meet the consumer demand of providing ever more recyclable content in consumer goods such as cosmetic compacts.

SUMMARY

(5) One aspect of the disclosure is directed to a tri-fold recyclable compact base. The tri-fold recyclable compact base also includes a pan receiving portion including an inner ring and an outer ring, the inner ring or the outer ring including at least one catch, and the inner ring configured to removably receive a cosmetics pan; a pan securing portion, the pan securing portion including a top flange and an inner ring extending perpendicularly from the top flange and an outer ring extending perpendicularly from the top flange, the inner ring or the outer ring including at least one circumferential ring configured to mate with the at least one catch; a first hinge rotatably connecting the pan receiving portion and the pan securing portion; a mirror receiving portion; and a second hinge rotatably connecting the mirror receiving portion and the pan securing portion.

(6) Implementations of this aspect of the disclosure may include one or more of the following features. The tri-fold recyclable compact base where rotation of the pan securing portion about the first hinge causes the at least one catch to receive the at least one circumferential ring and secure the pan securing portion to the pan receiving portion. The inner ring and outer ring of the pan receiving portion each include a catch. The inner ring and the outer ring each include a circumferential ring configured to mate with the catch of the inner ring and the outer ring of the pan receiving portion. Rotation of the pan securing portion about the first hinge causes the catches on each of the inner ring and the outer ring of the pan receiving portion to receive the circumferential ring on each of the inner ring and outer ring on the pan securing portion to secure the pan securing portion to the pan receiving portion. The tri-fold recyclable compact base further includes a pan of cosmetics in the pan receiving portion. The pan securing portion includes a lip extending from the top flange that overlaps the pan, the lip extending over the pan in the pan receiving portion. The tri-fold recyclable compact base further includes a cover secured in the mirror receiving portion. Tabs on the mirror receiving portion mate with a circumferential ring on the cover to secure the cover in the mirror receiving portion. The tri-fold recyclable compact base further includes a mirror inserted into the mirror receiving portion. The mirror is glued to the cover. The mirror is formed of a plastic and glued to the cover. The tri-fold recyclable compact base formed of polypropylene. The tri-fold recyclable compact base further includes an opening latch formed on the mirror receiving portion. The opening latch mates with the first hinge to hold the mirror receiving portion proximate the pan receiving portion. The mirror receiving portion includes a molded-in cover configured to have a mirror glued thereto.

(7) These and other aspects of the disclosure are described herein with respect to the following drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is an exploded view diagram of a compact in accordance with the disclosure;
- (2) FIG. 2 is a cross sectional view of a portion of the compact depicted in FIG. 1 in a closed state and in accordance with the disclosure;
- (3) FIG. 3A is a rear view of a compact in accordance with the disclosure depicting a portion of a hinge allowing for rotational movement of portions of the compact;
- (4) FIG. 3B is a left-side view or right-side view of a compact in accordance with the disclosure, the two sides are symmetrical and appear identical;
- (5) FIG. 3C is a top view of the compact of a compact in accordance with the disclosure;
- (6) FIG. 3D is a cross-sectional view of a compact in accordance with the disclosure along line A-A of FIG. 3C, extending through a hinge and a latch of the compact;
- (7) FIG. 3E is a bottom view of a compact in accordance with the disclosure;
- (8) FIG. 3F is a cross sectional of a compact in accordance with the disclosure along line B-B of FIG. 3C;
- (9) FIG. 3G is a front view of a compact in accordance with the disclosure depicting a latch mechanism to hold the compact in a closed position;
- (10) FIG. 3H is a left-side view or right-side view of a compact in accordance with the disclosure with the compact in a filled but open orientation, the two sides are symmetrical and appear identical; and
- (11) FIG. 3I is a perspective view of a compact in accordance with the disclosure in an open position.
- (12) FIG. 4A is an exploded view of a tri-fold base receiving the mirror and cover and the pan;
- (13) FIG. 4B is a perspective view of a compact as it might be shipped from a manufacturer to a cosmetic company, with the mirror receiving portion folded under the pan securing portion;
- (14) FIG. 4C is a perspective view of actions undertaken by a cosmetic company, with the unfolding of the mirror receiving portion from the pan securing portion;
- (15) FIG. 4D is a perspective view of the compact with the pan securing portion secured into the pan receiving portion and the compact in an open position; and
- (16) FIG. 4E is a perspective view of the compact in a closed position.
- (17) Those of ordinary skill in the art will recognize that FIGS. 3A-3I each depict alignment lines or cross-sectional indicator lines as are commonly used in drafting to ensure symmetry and alignment of the views or to inform the viewer of the origin of one or more views, these lines do not form a portion of this design or features of this disclosure.

DETAILED DESCRIPTION

(18) The instant disclosure is directed to a cosmetic compact and methods of forming a cosmetic compact to substantially increase the reusability and recyclability of the compact. FIG. 1 depicts an exploded view of a compact **10** in accordance with the disclosure. The compact **10** may be formed of for example, polypropylene, post-consumer polypropylene, or another plastic material that is highly recyclable yet also sufficiently durable to be reused multiple times without suffering damage. The compact **10** is formed of a tri-fold base **12**. The tri-fold base **12** includes three portions including a pan receiving portion **14** having a solid bottom **15** and configured to receive a pan **16**, for example an aluminum pan having a powdered make-up, and not allow the pan **16** to fall through the pan receiving portion **14**. The tri-fold base **12** also includes a pan securing portion **18** that folds on a hinge **20** to mate with the pan receiving portion **14** and secure the pan **16** in the pan

receiving portion **14** (See FIG. 2). The tri-fold base **12** includes a mirror receiving portion **22** configured to receive a mirror **24**. A removable cover **26** secures the mirror **24** in the mirror receiving portion **22**. The mirror receiving portion **22** is connected to the pan securing portion **18** by a hinge **27**. In accordance with one aspect of the disclosure the hinges described herein are formed of the same material as the remainder of the tri-fold base **12** and as part of the molding process. In this manner metal pins, as is often used for cosmetic compacts, are eliminated.

(19) In practice, the pan **16** is placed in the pan receiving portion **14**, but is not glued, rather it is received in an inner ring **28** formed in the bottom of the pan receiving portion **14**. The pan securing portion **18**, is connected to the pan receiving portion **14** via the hinge **20** (e.g., a living hinge), and upon folding of the pan securing portion **18** into the pan receiving portion **14**, the pan **16** is removably secured and prevented from falling out of the pan receiving portion **14**.

(20) Details of the pan securing ring **18** and the manner of securing the pan **16** can be seen in FIG. 2, which is a cross-sectional view of a portion of the compact **10**. The pan securing ring **18** includes an inner ring **30**, and outer ring **32**, and a top flange **34** spanning between the outer ring **32** and the inner ring **30** and a lip **35** extending over the pan **16**. The inner ring **30** includes a circumferential lip **36** configured to mate with a catch **38** formed in the inner ring **28** of the pan receiving portion **14**. A similar catch **40** is formed in an outer ring **42** of the pan receiving portion **14**, the catch **40** receives a circumferential lip **44** formed on the outer ring **32** of the pan securing portion **18**.

Though described as a catch **38**, **40** and circumferential lip **36**, **44** these are just exemplary means of securing the two components together in a releasable snap-fit configuration and other means may be employed without departing from the scope of the disclosure.

(21) The combination of the circumferential lip **36** and **44** with catches **38** and **40**, respectively, secures the pan receiving portion **14** and the pan securing portion **18** to one another and traps the pan **16** therebetween. This is not a permanent securing of the pan **16** in the pan receiving portion **14** as gluing of the pan **16** would be. In accordance with the disclosure, securing forces applied by the circumferential lips **36**, **44** and the catches **38**, **40** can be overcome by for example pulling on the mirror receiving portion **22** to cause the pan securing portion **18** to separate from the pan receiving portion **14**. The pan **16** may then be removed by the user and a new pan **16** (e.g., with fresh makeup) can be inserted. In this manner the volume of waste produced by the use of the compact **10** is reduced as the pan **16** may be replaced many times through the life of the compact **10**.

(22) In addition to the pan **16** being removable, application of pressure to the mirror **24** causes the mirror **24** and cover **26** releases from the mirror receiving portion **22**. The mirror **24** may then be separated from the cover **26**. The cover may be formed of the same material as the tri-fold base **12** and thus can be recycled together. The pan **16**, which may be formed of, for example, aluminum is highly recyclable as well, albeit separately from the cover **26** and tri-fold base **12**, which are formed of plastics, as noted above. The mirror **24** may be recycled, or if not, possible this forms the only non-recyclable waste associated with the compact **10**, greatly reducing the amount of waste as compared to current makeup container products. Further in some instances the mirror **24** may be glued to the cover **26**, and while not ideal still greatly reduces the volume of waste and enables recycling of the tri-fold base **12** and pan **16**.

(23) As noted above hinge **27** (e.g., a living hinge) connects the mirror receiving portion **22** to the pan securing portion **18**. The mirror receiving portion **22** includes tabs **46** configured to mate with a circumferential ring **48** formed on the cover **26**. When forced together in mating engagement the tabs **46** and circumferential ring **48** secure the mirror and more particularly the cover on which it is mounted in the mirror receiving portion **22**. The mirror **24** and cover **26** are secured in the mirror receiving portion **18** in normal use, but as noted above upon emptying of the cosmetics from the pan **16**, a user may place pressure against the mirror **24** and disengage the circumferential ring **48** on the cover **26** from the tabs **46** in the mirror receiving portion **22**, thus separating the mirror **24** and cover **26** from the tri-fold base **12**.

(24) FIGS. 3A-3I depict the detailed design of the compact **10**. FIG. 3A is a rear view of the

compact **10** showing a portion of hinge **27**. Line **50** notes the separation point of the mirror receiving portion **22** and the pan receiving portion **14**, when the compact **10** is in the closed position. FIG. **3B** is a side view of the compact **10**. As the compact is round the left and right-side views of the compact appear the same. FIG. **3C** is a top view of the compact **10** with the cover **26** secured in the mirror receiving portion **22**. FIG. **3D** is a cross sectional view of FIG. **3C** along line A-A. FIG. **3E** is a bottom view of the compact **10** with an opening latch **52** on one end and hinge **27** on the opposite side. As will be appreciated, application of pressure to the opening latch **52** opens the compact by rotating the mirror receiving portion **22** about the hinge **27**. FIG. **3F** is a cross-sectional view of the compact **10** along line B-B of FIG. **3C**. The detail section of this cross-sectional view is depicted in FIG. **2**. FIG. **3G** is a front view of the compact **10** showing the opening latch **52** and the mirror receiving portion **22** and the pan receiving portion **14**. FIG. **3H** depicts a side view of the compact **10** in the open position and FIG. **3I** depicts the compact **10** as a perspective view and in the open position.

(25) As will be appreciated, cosmetic companies typically do not manufacture the products they sell and particularly not the containers (e.g., compact **10**) in which their products are sold. The manufacturer may, in some instances, not only produce the compact **10** but may in some instances fill or partially fill the compact **10**. In accordance with one aspect of the disclosure, following manufacture of the tri-fold base **12**, the cover **26** and mirror **24** may be inserted into the mirror receiving portion **22** as shown in FIG. **4A**. Optionally, the pan **16** with the cosmetic may be inserted into the pan receiving portion **14** by the manufacturer of the tri-fold base. In other instances, the cosmetics company or a third party contractor may insert the pan **16** into the pan receiving portion **14** at their facilities.

(26) As shown in FIG. **4B** the mirror receiving portion **22** may be folded under the pan securing portion **18**. By folding the mirror receiving portion **22** under the pan securing portion **18** creates a low-profile subassembly that can be substantially flat packed to reduce the volume of the products being shipped (e.g., to a cosmetics company or third-party for filling with the cosmetic). The subassembly, with reduced profile may then be shipped to a customer such as a cosmetics company. However, those of ordinary skill in the art will understand that the disclosure is not so limited and the tri-fold base **12** may be shipped in any configuration, whether filled or not, from completely unfolded to completely folded without departing from the scope of the disclosure.

(27) The cosmetics company, or third-party filler, will upon receipt of the compact **10** unfold the mirror receiving portion **22** from the pan securing portion **14** as shown in FIG. **4C**. If necessary, they may then place a pan **16** with the cosmetics in the pan receiving portion **14** and fold via hinge **20** the pan securing portion **18** onto the pan receiving portion **14** to secure the pan **16** therebetween and result in an open compact **10** as depicted in FIG. **4D**. Finally, if needed an applicator or puff (not shown) may be inserted into the compact **10** and the mirror receiving portion **22** and cover **26** folded on hinge **27** to close the compact **10** as shown in FIG. **4E**. The completed and filled compact **10**, whether filled by the cosmetics company or the third-party filler, is then packaged and distributed through the standard channels of commerce and ultimately purchase by a user.

(28) The opening latch **52** formed on the perimeter of the mirror receiving portion **22** mates with the hinge **20** to releasably hold the compact **10** in the closed position as shown in FIG. **4E**. At any time, the user wishes to access the cosmetic, deflecting the latch away from the hinge **20** allows for opening of the compact **10** by hinged motion of the mirror receiving portion **22** via hinge **27** relative to the combination of the pan receiving portion **14** and the pan securing portion **18**, revealing both the cosmetic in the pan **16** and the mirror **24**.

(29) Upon completion of use, the user may easily disassemble the compact by first opening the compact **10** as described above. Once open, application of pressure to the mirror **24** and cover **26** forces the combination from the mirror receiving portion **22**. The mirror **24** and cover **26** may not be recyclable owing to the mirror **24** glued to the cover **26**. However, in some instances the mirror **24** may be separated from the cover **26**, thus further increasing the percentage of the total compact

that is recyclable. Still further, in some instances the mirror **24** may be made of a plastic material (e.g., a plastic resin) enabling the mirror **24** and the cover **26** to be recycled without requiring separation of the mirror **24** from the cover **26**.

(30) Next, application of pressure to the pan securing portion **18** can overcome the restraining force of the circumferential lips **36** and **44** with catches **38** and **40** and enable the pan securing portion **18** to rotate on the hinge **20** relative to the pan receiving portion **14**. Once in this position, substantially as shown in FIGS. **1**, application of pressure to an exterior surface of the pan receiving portion **16** overcomes any pressure applied to the pan by the circumferential lip **36** and **44** with catches **38** and **40** the pan **16** is forced free from the pan receiving portion **16**. Both the aluminum pan **16** and the tri-fold base **12** may now be separately recycled in their respective recycling streams. Additionally or alternatively, the pan **14** may also be returned to the cosmetic company or a third party, for refilling of the cosmetic. Further, the cosmetic company may sell pans **16** with the cosmetic therein for replacement in the tri-fold base **12**, allowing the user to replace just the cosmetic and re-using the tri-fold base **12**.

(31) In yet a further aspect of the disclosure, rather than having an insertable cover **26** and mirror **24**, the tri-fold base **12** may be formed such that the mirror receiving portion **22** is formed in a closed fashion, that is the cover is molded as part of the tri-fold base. In this scenario, the mirror **24** may be glued into the mirror receiving portion **22**. The molded in cover becomes to top of the compact **10** once assembled. The remainder of the tri-fold base **12** remains substantially the same and the pan **14** remains removable resulting in a refillable compact **10**.

(32) Still a further aspect of the disclosure is directed to a compact **10** employing a snap-fit cover (not shown). In this aspect of the disclosure, the pan receiving portion **14** receives the pan, and a pan securing portion **18** folds over the pan **16** and mates with the pan receiving portion **14** to secure the pan therein. A mirror receiving portion **22** may be separately formed or separated from the tri-fold base **12** (e.g., at hinge **27**) and trimmed to remove any sharp edges. The mirror receiving portion **22** may be formed with a solid back (i.e., integrally formed with the cover **26**). The mirror **24** may be glued into the mirror receiving portion. Using the snap-fit, the mirror receiving portion **18** can be completely removed from the combination of the pan receiving portion **14** and the pan securing portion **18** which retain the pan **16** and the cosmetics located therein.

(33) Though depicted in the drawings and described herein as being generally round in shape, the compact **10** is not so limited. Other shapes including square, rectangular, triangular, octagonal, etc., are contemplated within the many shapes of the compact without departing from the scope of the disclosure. Further though the pan receiving portion **14** is depicted as receiving a single pan **16**, the disclosure is not so limited and multiple pans **16** may be inserted into a pan receiving portion **14**. Further the pan receiving portion **14** may be configured with one or more ribs (not shown) which are configured to receive the multiple pans **16**. As an example, the pan receiving portion **14** may be square in shape and configured to receive a rectangular pan **16** and two square pans **16**, with two ribs formed in the bottom of the pan receiving portion **14** to receive and securely hold the three pans **16**.

(34) As a result of the use of a mono-material, such as polypropylene, and the tri-fold construction described herein, the amount of recyclable material is increased, and the volume of waste materials can be further reduced or eliminated.

(35) Still a further aspect of the disclosure is directed to a hot pour or direct pour cosmetic. A hot pour cosmetic is one that is poured directly into the pan receiving portion **14**. No pan **16** is needed for hot pour cosmetics. To recycle or reuse a tri-fold base **12** for use with hot pour cosmetics, any residual cosmetic need just be washed out of the pan receiving portion **14**. Further, in accordance with the disclosure, the initial use of the tri-fold base **12** may be to receive hot pour cosmetics, and then following an initial use, the user can wash the pan receiving portion **14** and insert a pan **16** having cosmetics therein to allow for reuse of the compact **10**.

(36) Although several embodiments and examples are disclosed herein, the present application

extends beyond the specifically disclosed embodiments to other alternative embodiments and/or uses of the disclosure and modifications and equivalents thereof. It is also contemplated that various combinations or sub-combinations of the specific features and aspects of the embodiments may be made and still fall within the scope of the disclosure. Accordingly, it should be understood that various features and aspects of the disclosed embodiments can be combined with or substituted for one another in order to form varying modes of the disclosed inventions. Thus, it is intended that the scope of the present disclosure herein disclosed should not be limited by the particular disclosed embodiments described above.

Claims

1. A tri-fold recyclable compact base comprising: a pan receiving portion including an inner ring and an outer ring, the inner ring or the outer ring including at least one catch, and the inner ring configured to removably receive a cosmetics pan; a pan securing portion, the pan securing portion including a top flange and an inner ring extending perpendicularly from the top flange and an outer ring extending perpendicularly from the top flange, the inner ring or the outer ring including at least one circumferential lip configured to mate with the at least one catch; a first hinge rotatably connecting the pan receiving portion and the pan securing portion; a mirror receiving portion; and a second hinge rotatably connecting the mirror receiving portion and the pan securing portion.
2. The tri-fold recyclable compact base of claim 1, wherein rotation of the pan securing portion about the first hinge causes the at least one catch to receive the at least one circumferential lip and secure the pan securing portion to the pan receiving portion.
3. The tri-fold recyclable compact base of claim 1, wherein the inner ring and the outer ring of the pan receiving portion each include a catch.
4. The tri-fold recyclable compact base of claim 3, wherein the inner ring and the outer ring each include a circumferential lip configured to mate with the catch of the inner ring and the outer ring of the pan receiving portion.
5. The tri-fold recyclable compact base of claim 4, wherein rotation of the pan securing portion about the first hinge causes the catches on each of the inner ring and the outer ring of the pan receiving portion to receive the circumferential lip on each of the inner ring and outer ring on the pan securing portion to secure the pan securing portion to the pan receiving portion.
6. The tri-fold recyclable compact base of claim 1, further comprising a pan of cosmetics in the pan receiving portion.
7. The tri-fold recyclable compact base of claim 6, wherein the pan securing portion includes a lip extending from the top flange that overlaps the pan, the lip extending over the pan in the pan receiving portion.
8. The tri-fold recyclable compact base of claim 1, further comprising a cover secured in the mirror receiving portion.
9. The tri-fold recyclable compact base of claim 8, wherein tabs on the mirror receiving portion mate with a circumferential ring on the cover to secure the cover in the mirror receiving portion.
10. The tri-fold recyclable compact base of claim 9, further comprising a mirror inserted into the mirror receiving portion.
11. The tri-fold recyclable compact base of claim 10, wherein the mirror is glued to the cover.
12. The tri-fold recyclable compact base of claim 10, wherein the mirror is formed of a plastic and glued to the cover.
13. The tri-fold recyclable compact base of claim 1, formed of polypropylene.
14. The tri-fold recyclable compact base of claim 1, further comprising an opening latch formed on the mirror receiving portion.
15. The tri-fold recyclable compact base of claim 14, wherein the opening latch mates with the first hinge to hold the mirror receiving portion proximate the pan receiving portion.

16. The tri-fold recyclable compact base of claim 1, wherein the mirror receiving portion includes a molded-in cover configured to have a mirror glued thereto.
 17. The tri-fold recyclable compact base of claim 1, wherein at least one of the first hinge and the second hinge are living hinges.
 18. The tri-fold recyclable compact base of claim 1, wherein the pan receiving portion is configured to receive multiple pans.
 19. The tri-fold recyclable compact base of claim 1, wherein the each of the pan receiving portion, the pan securing portion, and the mirror receiving portion have a round shape.
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